Name: Aditi Rajak

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Section: B2

```
sit@sit-Vostro-3400:~/cclab$ yacc -d exp9.y
sit@sit-Vostro-3400:~/cclab$ flex exp9.l
sit@sit-Vostro-3400:~/cclab$ gcc y.tab.c lex.yy.c -o exp9 -ll -lm
sit@sit-Vostro-3400:~/cclab$ ./exp9
Enter an expression:
2^3
Result: 8
5+6
Result: 11
8++
Result: 9
9--
Result: 8
7%4
Result: 3
```

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```
%{
#include <stdio.h>
#include <stdlib.h>
#include "y.tab.h"
%}

%%

[0-9]+      { yylval = atoi(yytext); return NUMBER; }
"++"        { return INC; }
"--"        { return DEC; }
[ \t]       ;
\n          { return '\n'; }
[+\-*/^%()] { return yytext[0]; }
.           { return yytext[0]; }

%%

int yywrap(void)
{
    return 1;
}
```

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exp9.y

```
%{
#include <stdio.h>
#include <stdlib.h>
#include <math.h>

int yylex(void);
int yyerror(const char *s);
%}

%token NUMBER INC DEC
%left '+' '-'
%left '*' '/' '%'
%right '^'
%right INC DEC

%%

input:
    | input line
    ;

line:
      exp '\n'  { printf("Result: %d\n", $1); }
    | '\n'
    ;

exp:
      NUMBER        { $$ = $1; }
    | exp '+' exp   { $$ = $1 + $3; }
    | exp '-' exp   { $$ = $1 - $3; }
    | exp '*' exp   { $$ = $1 * $3; }
    | exp '/' exp   { $$ = $1 / $3; }
    | exp '%' exp   { $$ = $1 % $3; }
    | exp '^' exp   { $$ = (int)pow($1, $3); }
    | exp INC       { $$ = $1 + 1; }
    | exp DEC       { $$ = $1 - 1; }
    | '(' exp ')'   { $$ = $2; }
    ;

%%

int main(void)
{
    printf("Enter an expression:\n");
    yyparse();
    return 0;
}

int yyerror(const char *s)
{
    printf("Invalid expression\n");
    return 0;
}
```